

RTFDD: A Robust Two-Stage Framework For Web Defacement Detection

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Abstract. Defacement attacks currently pose one of the most significant threats to web applications. Attackers often utilize this method to leave messages and signatures, which serve as a form of propaganda or a demonstration of power. However, existing detection methods mainly focus on content-based analysis, which can consume substantial computational resources, require extended processing times, and reduce accuracy. Furthermore, these methods typically overlook the attackers’ signatures—an indicator that may seem simple but is highly effective for detection. This paper proposes a defacement detection framework that integrates signature analysis with an attack detection model, combining image, text, and HTML features. This framework is capable of effectively handling both static and dynamic web pages. Experimental evaluations conducted on a dataset of 20,255 collected signatures and 96,220 web pages demonstrate that our approach achieves an accuracy rate of up to 99.23%, with a False Alarm Rate (FAR) of only 0.38%, surpassing existing detection methods.

1 Introduction

Web defacement is a type of attack targeting web servers, typically involving alterations to the content, layout, color scheme, or images of a website [1,2]. The primary purpose of such attacks is to highlight security vulnerabilities to the website owner. However, certain groups also exploit defacement attacks to sabotage websites, damage their reputation, or deceive users by spreading false or misleading information. Hackers commonly deface websites to leave behind their signatures and messages, resembling street art in the digital realm. Many attacker groups use these defaced web pages as a canvas to showcase their presence and achievements, often as a form of self-promotion or ideological expression.

Web defacement detection techniques are commonly divided into anomaly-based, signature-based, and Machine Learning (ML) approaches. Traditional

anomaly-based methods, such as checksum and DIFF comparison, are simple and fast but only suitable for static websites. In contrast, advanced techniques like statistical analysis, genetic programming, and ML handle dynamic websites more effectively but require higher computational resources. Signature-based detection relies on known attack patterns for quick identification but cannot detect novel attacks. Despite extensive research, few solutions combine the speed of signature-based methods with the adaptability of anomaly-based techniques.

This section reviews prior studies on both approaches. Signature-based methods rely on predefined patterns derived from previously known attacks, such as specific keywords, URLs, or HTML structures. These techniques enable fast and accurate detection of previously observed attacks but face challenges in identifying new or modified attack variants. Harahap *et al.* [3] developed a low-computational-cost method for detecting website defacement in university domains using predefined attack signatures. Their approach utilizes patterns such as keywords, URLs, or HTML structures. However, its dependence on known signatures and limited capability in analyzing dynamic content or multimedia elements reduces its effectiveness against novel attacks. To overcome these limitations, hybrid techniques have been explored. Hoang *et al.* [4] proposed a system that combines supervised ML with signature-based integrity verification using hash functions. Although effective in detecting tampering in static HTML content, the hash-based approach is sensitive to minor changes, which may lead to a high false positive rate when applied to dynamic or multimedia content.

Anomaly-based detection models, particularly those utilizing ML, have attracted considerable research interest. Burruss *et al.* [5] used a finite mixture model to classify attackers into low- and high-volume groups, offering behavioral insights but lacking real-time detection capabilities. Almomani *et al.* [6] proposed an Enhanced URL Defacement Detection model using Particle Swarm Optimization (PSO) to improve feature selection, achieving 92.5% accuracy on static datasets. However, its reliance on URL features limits performance on dynamic websites and multimedia content. Wu *et al.* [7] applied a Support Vector Machine (SVM) to HTML features, reaching 87.3% accuracy but meeting with high FAR due to fixed thresholds and weak performance on visually rich content. Hoang *et al.* [8] employed BiLSTM on webpage text yet ignored HTML structure and embedded media, reducing effectiveness on complex pages. Nguyen *et al.* [9] combined text and screenshot data but only averaged outputs from separate models without integrating cross-modal interactions, limiting overall performance.

The limitations of current website defacement detection solutions include (1) focusing mainly on URL content while neglecting actual webpage content, leading to low effectiveness; (2) being suitable only for static websites; (3) high false alarm rates due to fixed detection thresholds; (4) inadequate processing of images and HTML structure; and (5) limited feature integration, relying only on separate evaluations of individual data types. To address these issues, this paper makes three key contributions as follows:

- the construction of a comprehensive dataset containing over 96,000 web pages with image, text, and HTML data, along with more than 20,200 unique defacement signatures for improved signature-based detection;
- the proposal of a multimodal detection model that combines image, text, and HTML features, outperforming prior anomaly-based methods;
- the development of a two-stage detection framework that integrates signature-based and multimodal analysis to enhance accuracy, reduce false positives, and improve processing speed.

The rest of this paper is organized as follows: Section 2 introduces the proposed approach, Section 3 presents and analyzes the experimental results. In Section 4, we have a conclusion and suggest the future direction of this study.

2 Proposed Methodologies

This section introduces the proposed robust two-stage framework for the web defacement detection model, RTFDD. The model combines signature-based detection in the first stage with a deep learning-based detection approach in the second stage. If no known signature is detected during the initial phase, the system proceeds to the deep learning model, which leverages multimodal features - including visual, textual, and HTML characteristics - to identify potential defacement attacks.

2.1 Overview

Figure 1 illustrates the overall architecture of the proposed RTFDD (Robust Two-Stage Framework for Web Defacement Detection). The hybrid framework consists of two main stages: a signature-based detection module and an anomaly-based deep learning module. In the first stage, incoming webpages are first processed and compared against a database of predefined defacement signatures. If a match is found, the system immediately raises a defacement alert. If no match is detected, the input proceeds to the second stage, where a multimodal deep learning model is used for anomaly detection. In the anomaly-based stage, the webpage content is divided into three modalities: HTML structure, textual content, and screenshot image. Each modality is processed separately through dedicated modules: HTML features are extracted using a pretrained BERT model applied to the DOM tree, text features are processed through a text encoder, and image features are extracted via a CNN-based model. These features are then fused through a transformer-based integration layer, followed by a multi-layer perceptron classifier to make the final defacement decision.

2.2 Signature-based Attack Detection

The first stage of the RTFDD framework leverages a signature-based detection mechanism to rapidly identify known web defacement attacks. This stage is designed to be lightweight and computationally efficient, making it suitable for

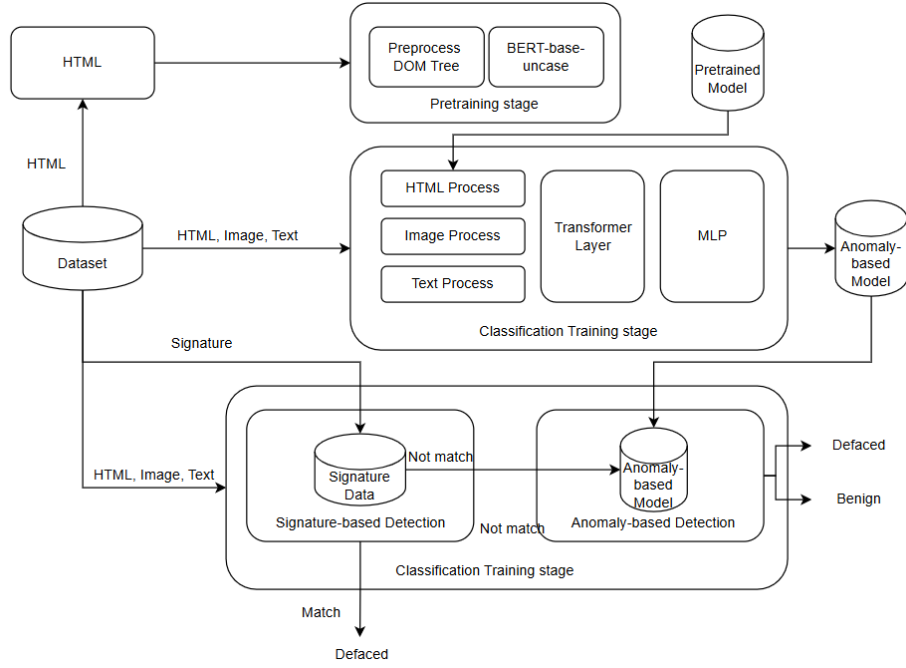


Fig. 1. The proposed two-stage framework for web defacement detection.

real-time monitoring of large-scale web environments. To implement this mechanism, a comprehensive signature database has been constructed, consisting of 20,255 unique signatures extracted from previously recorded web defacement incidents. These signatures were collected from publicly available defacement archives, security forums, and manually labeled datasets. Each signature encodes distinct indicators of compromise, including specific keywords commonly used by attackers (e.g., "Hacked by", "Owned by"). During detection, the incoming webpage is scanned and tokenized to extract relevant textual and structural features. These features are then matched against the signature database using a fast matching algorithm based on approximate string matching and rule-based pattern recognition. If a match is found, the system immediately classifies the page as defaced, and an alert is generated. This early detection step allows the system to block well-known attacks quickly, reducing the burden on the more resource-intensive deep learning model in the second stage.

2.3 Anomaly-based Attack Detection Using Feature Fusion

In the second stage of the RTFDD framework, an anomaly-based detection approach is employed to identify web defacement attacks that cannot be captured by known signatures. This stage utilizes a multimodal deep learning architecture

that integrates features extracted from the HTML structure, textual content, and webpage screenshots.

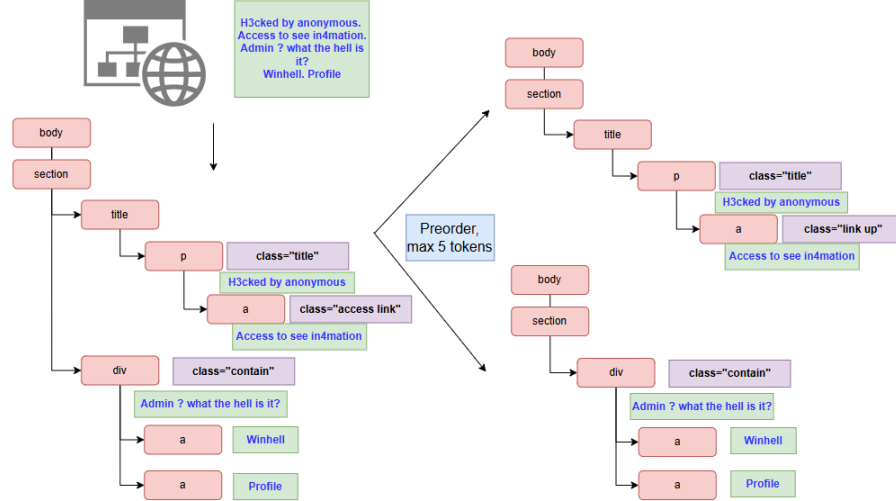


Fig. 2. DOM Tree Processing with Pre-order and Max 5 tokens.

HTML Feature Extraction: The HTML content of each webpage is first parsed into a DOM tree, where each node contains both tag information and associated textual content. To reduce noise, non-informative elements such as `<script>` and `<style>` tags are removed, while useful attributes like `class` and `id` are retained. Given the complexity and size of DOM trees, the structure is then divided into overlapping subtrees using a sliding window based on a fixed number of tokens. This preserves local context and prevents the loss of important structural relationships.

Each subtree is serialized and encoded using a transformer-based encoder with modified positional embeddings to reflect parent-child and sibling relationships between nodes. In particular, despite the initial cleaning process, many DOM trees still consist of thousands of tokens and data points, many of which lack semantic significance. This poses a challenge when feeding the data into modern deep learning models. To address this, the DOM tree is further divided into overlapping subtrees using a sliding window approach based on a fixed token size m . Unlike conventional text-based sliding windows, which often disrupt crucial parent-child relationships, this method keeps semantically related nodes intact. The segmentation process begins at the root of the DOM tree and follows a pre-order traversal. The first subtree is constructed by collecting up to m tokens. Then, additional subtrees are created by gradually expanding the current subtree with newly encountered nodes, while older, less relevant nodes are pruned. This dynamic expansion and pruning mechanism help retain meaningful

structural context throughout the segmentation. For instance, as illustrated in Figure 2, the tree is traversed in pre-order with a maximum token limit of 5. At each iteration, the subtree is updated by incorporating new nodes and discarding those that fall outside the sliding window. The final subtree at any point contains the most recently added nodes along with relevant nodes preserved from earlier steps.

The model is further pretrained based on Bert-base-uncased [10] on unlabeled HTML webpages using the Masked Language Modeling (MLM) objective to learn structural and semantic representations specific to web layout. The core of this encoder is the self-attention mechanism [11], where the model learns to create representations of a token by aggregating information from other tokens in the input sequence. Specifically, this paper first randomly selects 15% of token positions, then repeatedly samples DOM nodes and adds all their tokens to the mask until the total number of masked tokens reaches the desired 15%. As in BERT, for a chosen position, (1) replace it with a special token [MASK] 80% of the time, (2) replace it with a random token 10% of the time, and (3) leave it unchanged in the remaining cases as described in Figure 3.

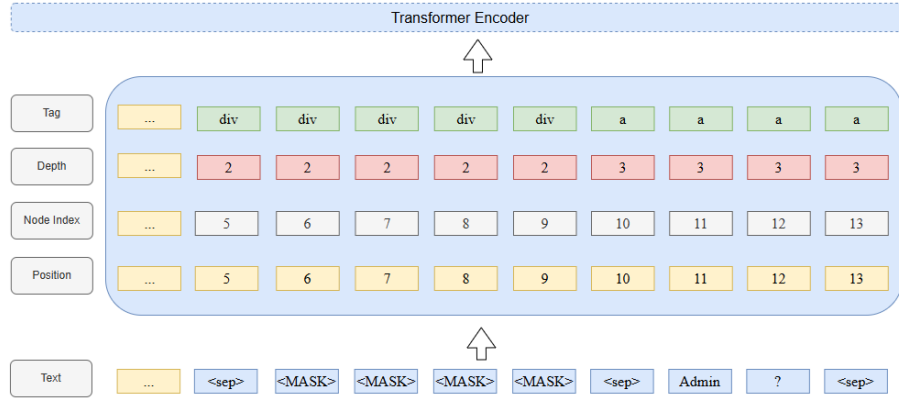


Fig. 3. Pretraining with HTML.

Text and Image Feature Extraction: Textual data extracted from the webpage body is preprocessed and tokenized by Tokenizer [12]. A Bidirectional Long Short-Term Memory (BiLSTM) network is used to encode the sequence of tokens, capturing both forward and backward contextual dependencies. The output is a dense vector representation reflecting the semantic structure of the text content. Webpage screenshots are standardized to a fixed resolution of 224×224 pixels, and their pixel values are normalized to the range $[0, 1]$. The preprocessed images are then passed through an EfficientNetB0 [13] to extract visual features that capture layout, color distribution, and embedded visual elements relevant to potential defacement indicators.

Fusion Feature: The feature vectors obtained from the HTML (denoted as H), text (T), and image (I) branches are concatenated into a unified set of feature vectors: $F = H \cup T \cup I$. These vectors are passed into a transformer encoder with self-attention, which allows the model to compute interactions across different modalities, learning joint representations that integrate structural, textual, and visual cues. The output of the transformer is then fed into a multi-layer perceptron (MLP) classifier, which outputs a binary label indicating whether the webpage has been defaced or is benign. This multimodal design addresses key limitations of previous methods by enabling the model to detect sophisticated defacement attempts that manipulate layout, alter textual content, or modify visual presentation. It ensures robustness across both static and dynamic webpages and enhances generalization to previously unseen attack patterns.

3 Experiments and Evaluation

3.1 Dataset Processing

Previous studies in this domain have highlighted several limitations, such as the use of statistical methods without access to appropriate datasets, reliance on limited or confidential data, and a narrow focus on specific modalities such as text or image, which reduces detection effectiveness. Moreover, prior research has not introduced any comprehensive signature datasets specifically designed for defacement detection. To ensure a fair and reliable evaluation, we re-implemented representative detection methods and developed a customized dataset for training and testing purposes. This dataset includes both normal and defaced websites: normal samples were collected from top-ranked Alexa websites [14], while defaced samples were obtained from Zone-h.org [15]. Using a toolset based on Selenium WebDriver [16] and BeautifulSoup, we extracted HTML content, textual information, and page screenshots from each web page. The final dataset comprises 57,120 labeled normal samples and 39,100 defaced samples. From the 39,100 defaced samples, we further extracted a collection of 20,255 distinct defacement signatures, which serve as the foundation for the signature-based detection component of our system.

3.2 Experimental Setups

For the pretraining phase, the model utilizes BERT-base-uncased as the backbone and performs fine-tuning with a hidden layer size of 768. The model is pretrained using unlabeled HTML data extracted from the prepared dataset for 5 epochs, with a batch size of 16 and a learning rate of $2e^{-5}$. For the classification phase, a deep learning-based model is employed. The training parameters of the classification model are summarized in Table 1. Notably, the final layers of the network consist of fully connected layers, with 256 units in the first layer, followed by 2 output units representing the two classes: defaced and benign.

Table 1. Experiment Setups

Component	Specification
CPU	CPU Intel Xeon (2.6 GHz)
GPU	NVIDIA Tesla P100 (VRAM 16GB)
RAM	29 GB
Epochs	10
Batch Size	16
Optimizer	Adam
Learning rate	$2e^{-5}$
Loss function	Cross Entropy

3.3 Results and Analysis

For the signature-based classification phase, the model utilizes the collected signature dataset. It performs pattern matching between the signatures and the input web content to determine whether the page has been defaced. Pages flagged as suspicious are then passed to the deep learning-based detection model for further verification.

Figure 4 illustrates comparable accuracy trends, highlighting the robustness of our deep learning-based model. The plotted curves demonstrate a consistent pattern, featuring rapid convergence in the initial stages and steady improvement thereafter. The close correlation between training and validation results reinforces the model’s generalization ability and its effectiveness in accurate detection.

For the deep learning-based classification phase, the paper utilizes the constructed dataset as described in Section 3.1, with a train-test split ratio of 80:20. The study evaluates several previously proposed methods on this newly constructed dataset, including Random Forest [4], BiLSTM [8], and BiLSTM + EfficientNet [9]; as well as the method by Wu *et al.* [7]. In addition, the paper evaluates two proposed deep learning-based detection models: one based on BERT and another based on feature fusion. Furthermore, the paper assesses the proposed two-stage RTFDD framework, which combines signature-based detection with the deep learning-based feature fusion model. The results are evaluated based on two main criteria: detection accuracy, including related metrics such as Recall, Precision, and F1-score, and execution performance, which includes inference time and detection speed. Since the proposed model incorporates a signature-based detection phase, the detection time is specifically measured on the 20% test subset to reflect realistic performance under deployment conditions.

Table 2 presents the experimental results of various defacement detection models, highlighting the effectiveness of RTFDD in identifying defacement attacks. The deep learning-based model utilizes a feature fusion strategy, combining HTML features extracted using the proposed method, textual features processed by a BiLSTM model, and visual features from the EfficientNetB0 model.

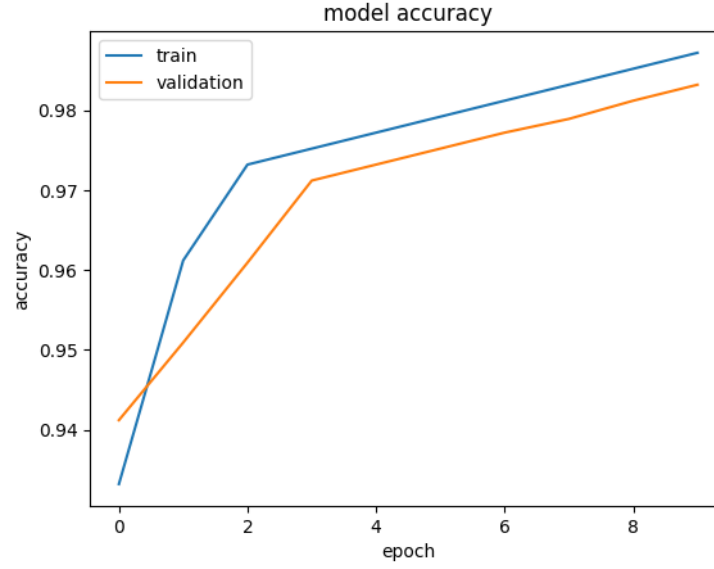


Fig. 4. Training and validating accuracy over epochs during the model training process

Table 2. Performance Comparison of Various Models

Model	Accuracy (%)	Precision	Recall	F1-score	Time (s)
Hoang <i>et al.</i> [4]	86.03	94.28	69.35	79.92	0.0007
Wu <i>et al.</i> [7]	91.28	91.26	91.28	91.26	0.0003
Hoang <i>et al.</i> [8]	96.54	96.93	94.43	95.66	0.0008
Nguyen <i>et al.</i> [9]	97.49	97.76	95.99	96.87	0.0112
BERT-base-uncased	90.56	93.50	82.63	87.73	0.0061
Pretrained only (Ours)	95.79	94.81	93.59	94.20	0.0062
Fusion (Ours)	98.97	98.76	98.39	98.57	0.0134
RTFDD (Ours)	99.23	99.62	98.14	98.87	0.0033

In addition, RTFDD integrates signature-based analysis with the proposed deep learning approach to enhance its ability to detect a wide range of defacement patterns. This hybrid mechanism allows the system to leverage both precise pattern matching and robust feature learning. As a result, RTFDD outperforms all previously proposed methods, achieving an impressive accuracy of 99.23% and an F1-score of 98.87%, while maintaining a low false alarm rate of just 0.38%. These results underscore the superior performance of RTFDD in detecting web defacement attacks. Notably, this combination also leads to significantly faster execution time compared to previous deep learning-based methods such as those proposed by Nguyen *et al.* [9] and others based only on deep learning models. Moreover, it delivers acceptable processing times relative to traditional machine learning approaches while offering substantially superior accuracy.

4 Conclusion

In this study, the authors proposed a new method for detecting web defacement attacks based on a two-stage model named RTFDD, which combines signature-based analysis with deep learning techniques that integrate visual, textual, and HTML features. An effective framework was introduced for extracting features from HTML, a data source that has been underutilized in previous research. During the process, the authors also constructed a comprehensive dataset comprising over 20,200 defacement signatures and more than 96,000 webpages, each with associated screenshots, textual, and HTML data. Experimental evaluation conducted on a dataset consisting of 20,225 signatures and 96,220 webpages confirmed the superior performance of the proposed method, achieving an accuracy of 99.23% and a false positive rate of only 0.38%. This detection approach enhances accuracy while maintaining fast computational performance, due to the incorporation of signature matching in the detection process. However, the deep learning component remains resource-intensive, which may limit the scalability of the model in certain real-world web security deployments. Future research directions include optimizing the deep learning module to minimize computational overhead, thereby facilitating deployment in resource-constrained or real-time web monitoring environments. Furthermore, extending the dataset to encompass multilingual content and a wider variety of website architectures is expected to enhance the generalizability and resilience of the proposed approach against emerging and sophisticated web defacement techniques.

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